

A close-up photograph of several darts hitting a target. The target is partially visible on the right side, showing its segmented design. The darts are in the foreground, with their flights and barrels clearly visible. A colorful heatmap overlay is applied to the target area, showing a concentration of hits in the center, represented by a bright yellow and orange spot, with the intensity fading to blue and purple towards the edges. The background is a solid dark blue.

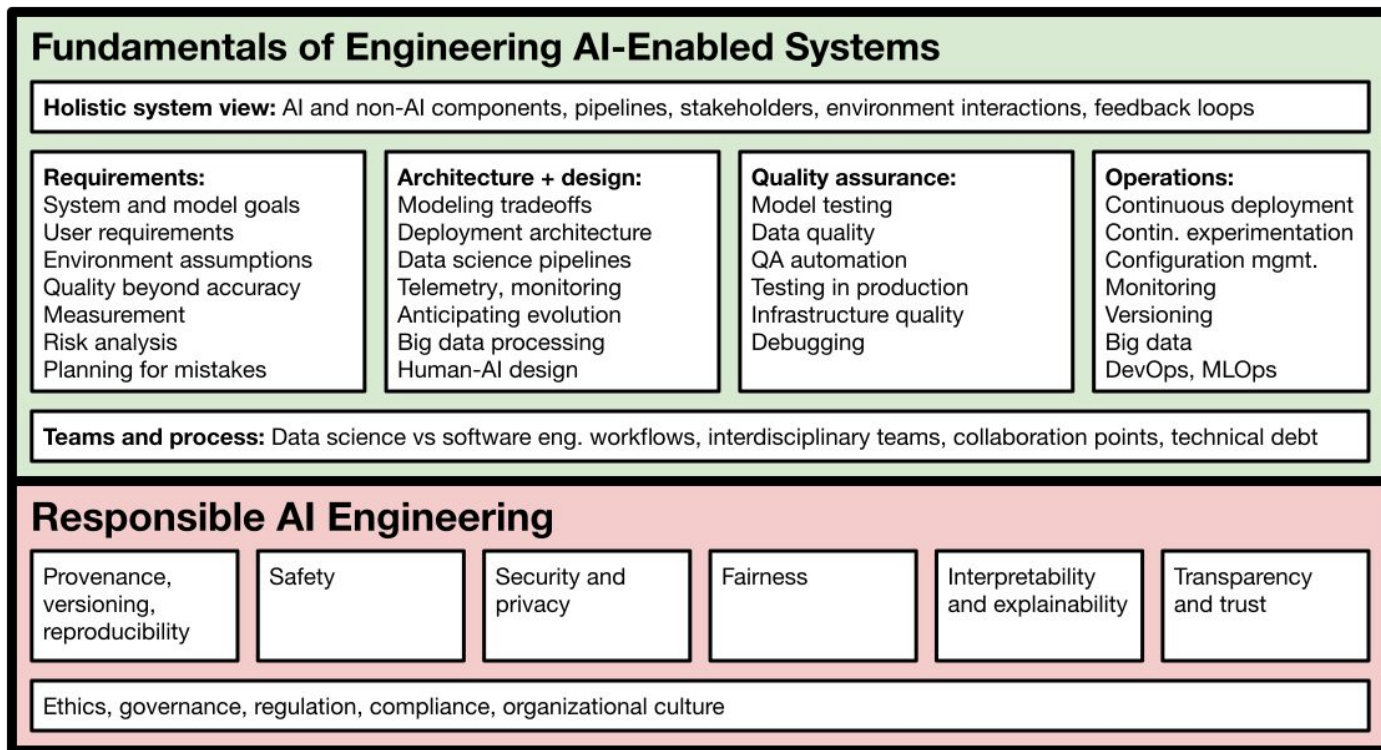
Machine Learning in Production

Model Quality: Accuracy, Pitfalls, and Measurement

Administrativa

- Fill out team survey (catme.org)

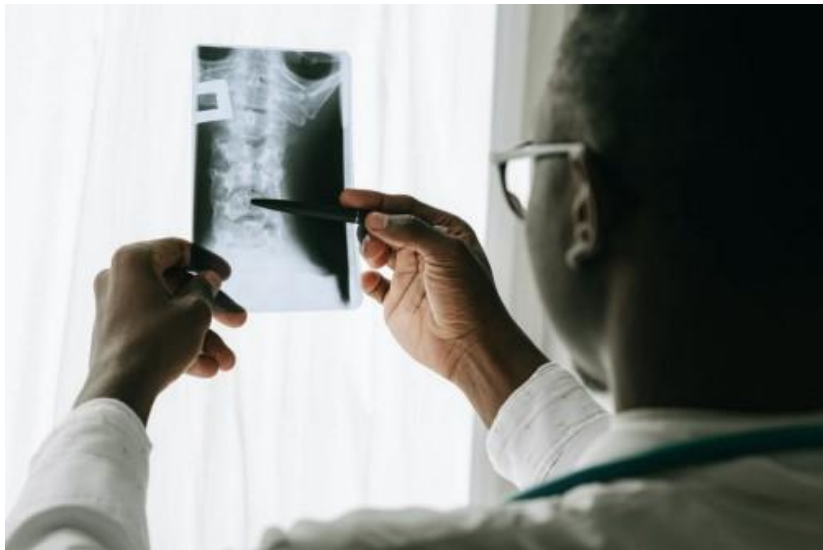
Jumping to QA...



Learning Goals

- Select a suitable metric to evaluate prediction accuracy of a model and to compare multiple models
- Select a suitable baseline when evaluating model accuracy
- Know and avoid common pitfalls in evaluating model accuracy
- Explain how software testing differs from measuring prediction accuracy of a model

Case Study: Cancer Prognosis



We should stop training radiologists now. It's just completely obvious that within five years, deep learning is going to do better than radiologists. --

[Geoffrey Hinton](#), 2016

The Model is Part of a System in an Environment

Tryton - Administrator - GNU SOLIDARIO HOSPITAL [Euro]

File User Options Favorites Help

screen

- Addresses
- Categories
- Product
- Financial
- Currency
- Inventory & Stock
- Purchase
- Calendar
- Health
- Patients**
- Institutions
- Appointments
- Prescriptions
- Demographics
- Laboratory
- Imaging
- Hospitalizations
- Surgeries
- Pediatrics
- Archives
- Nursing
- Health Services
- Reporting
- Configuration

Patients **Obstetric Hist ...**

Patients 1 / 8

New Save Switch Reload Previous Next Attachment(0) Action Relate Report E-Mail Print

Main Info

Betz, Ana Female Age: 29y 3m 20d

Critical Information

Personal history of allergy to penicillin
Insulin-dependent diabetes mellitus

Severe allergic reactions to β -lactams

General Info Socioeconomics Medication Diseases Surgeries Genetics Lifestyle QB/GYN

General Screening

Fertile: ☒ Pregnant: ☐ Menarche age: 12 Menopausal: ☐ Menopause age:

OB summary

Pregnancies: 1 Premature: 0 Abortions: 0 Stillbirths: 0

Menstrual History

Date	LMP	Length	frequency	volume	Regular	Dysmenorrhea	Reviewed	Institution
01/24/2015	01/20/2015	5	eumenorrhea	normal	☐	☐	Cordara, Cameron	GNU SOLIDARIO HOSPITAL

tryton://health.gnusolidario.org:8000/health28rc1/model/gnuhealth.patient/1;views=%5B223%2C+224%5D

Scope: Model Quality, not ML Algorithm/Data/System Quality

Today's focus is on the quality of the produced *model*, not the algorithm used to learn the model or the data used to train the model or the product around the model

- ➤ assuming *Decision Tree Algorithm* and feature extraction are correctly implemented (according to specification), is the model learned from data any good?

Focus on measuring quality, not debugging the source of quality problems

Out of Scope Today: System-Level Considerations

- Models used by radiologists, humans in the loop
- Radiologists are specialists who do not directly see patients
- Radiologists may not trust model, but are also overworked
- Radiologist must explain findings
- Patient can see findings before physician (CURES act)

Out of Scope Today: Model Qualities beyond Accuracy

But many *model qualities* matters when building a system beyond just *prediction accuracy*, e.g.:

- Model size
- Inference latency
- Learning latency
- User interaction model
- Ability to incrementally learn
- Explainability
- Robustness

Today and Next Week

Narrow focus on prediction accuracy of the model

That's difficult enough for now.

More on system vs model goals and other model qualities later

On Terminology



Model: $X \rightarrow Y$

Performance: In machine learning, "performance" typically refers to accuracy: "this model performs better" = it produces more accurate results

Be aware of ambiguity across communities (see also: performance in arts, job performance, company performance, performance test (bar exam) in law, software/hardware/network performance)

- When speaking of "**time**", be explicit: "learning time", "inference latency", ...
- When speaking of model **accuracy** use "prediction accuracy", ...

Part 1:

Measuring Prediction Accuracy for Classification Tasks

Typical Questions

Compare two models (same or different implementation/learning technology) for the same task:

- Which one supports the system goals better?
- Which one makes fewer important mistakes?
- Which one is easier to operate?
- Which one is better overall?
- Is either one good enough?

Concept check: Accuracy, Confusion Matrix, Precision, Recall, F1, False Positive/False Negative, AUC, Baseline?



Confusion/Error Matrix

$$accuracy = \frac{\text{correct predictions}}{\text{all predictions}}$$

```
def accuracy(model, xs, ys):  
    count = length(xs)  
    countCorrect = 0  
    for i in 1..count:  
        predicted = model(xs[i])  
        if predicted == ys[i]:  
            countCorrect += 1  
    return countCorrect / count
```

	Actually Grade 5 Cancer	Actually Grade 3 Cancer	Actually Benign
Model predicts Grade 5 Cancer	10	6	2
Model predicts Grade 3 Cancer	3	24	10
Model predicts Benign	5	22	82

$$\text{Example's accuracy} = \frac{10+24+82}{10+6+2+3+24+10+5+22+82} = .707$$

Is 99% Accuracy good?



Is 99% Accuracy good?

➤ depends on problem; can be excellent, good, mediocre, terrible

10% accuracy can be good on some tasks (information retrieval)

Always compare to a base rate!

$$\text{Reduction in error} = \frac{(1 - \text{accuracy}_{\text{baseline}}) - (1 - \text{accuracy}_f)}{1 - \text{accuracy}_{\text{baseline}}}$$

- from 99.9% to 99.99% accuracy = 90% reduction in error
- from 50% to 75% accuracy = 50% reduction in error

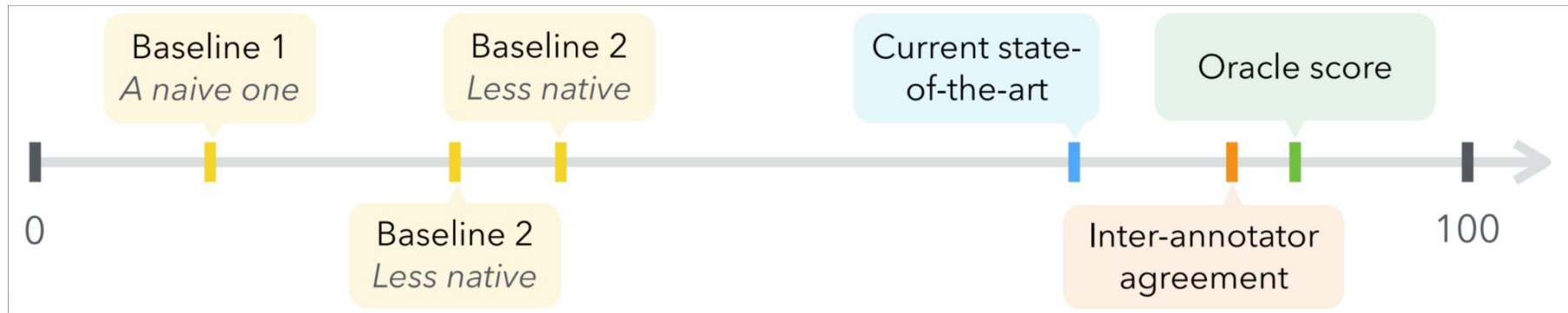
Baselines?

Suitable baselines for cancer prognosis? For audit risk prediction?



Consider Different Forms of Baselines

- Lower bound – result of a baseline solution (less complex or trivial system the performance of which is supposed to be easily surpassed)
 - Random-choice baseline
 - Most-frequent-value baseline
 - Simple heuristic
- Upper bound – oracle experiment result, human judgements, etc.
 - Oracle: an imaginary entity that makes the best possible decisions, given practical limitations.



Most-frequent-value Baseline for Imbalanced Classes

Predicting unlikely events -- 1 in 2000 has cancer ([stats](#))

Random predictor

	Cancer	No c.
Cancer pred.	3	4998
No cancer pred.	2	4997

.5 accuracy

Never cancer predictor

	Cancer	No c.
Cancer pred.	0	0
No cancer pred.	5	9995

.999 accuracy

See also [Bayesian statistics](#)

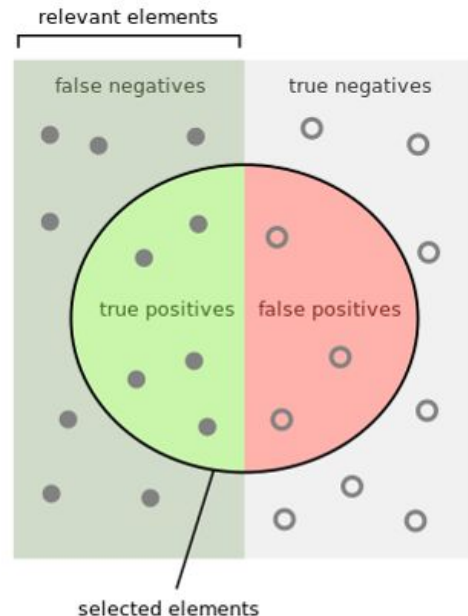
Besides Accuracy: Precision, Recall, F1

Measuring success of correct classifications (or missing results):

- Recall = $TP/(TP+FN)$
 - aka true positive rate, hit rate, sensitivity; *higher is better*
- False negative rate = $FN/(TP+FN) = 1 - \text{recall}$
 - aka miss rate; *lower is better*

Measuring rate of false classifications (or noise):

- Precision = $TP/(TP+FP)$
 - aka positive predictive value; *higher is better*
- False positive rate = $FP/(FP+TN)$
 - aka fall-out; *lower is better*
- Combined measure (harmonic mean): F1 score = $2 \frac{\text{recall} * \text{precision}}{\text{recall} + \text{precision}}$



(CC BY-SA 4.0 by [Walber](#))

False pos. and false neg. equally bad?

Accuracy does not take into account the differences in the misclassification costs for each class. Misclassifying a patient with a serious disease as having a benign condition is more costly.

- Suggesting products to buy on e-commerce site
- Identifying human trafficking at the border
- Predicting high demand for ride sharing services
- Predicting recidivism chance
- Approving loan applications

No answer vs wrong answer?

(This requires understanding the larger system!)

Extreme Classifiers

- Identifies every instance as negative (e.g., no cancer):
 - 0% recall (finds none of the cancer cases)
 - 100% false negative rate (misses all actual cancer cases)
 - undefined precision (no false predictions, but no predictions at all)
 - 0% false positive rate (never reports false cancer warnings)
- Identifies every instance as positive (e.g., has cancer):
 - 100% recall (finds all instances of cancer)
 - 0% false negative rate (does not miss any cancer cases)
 - low precision (also reports cancer for all noncancer cases)
 - 100% false positive rate (all noncancer cases reported as warnings)

Consider the Baseline Probability

Predicting unlikely events -- 1 in 2000 has cancer ([stats](#))

Random predictor

	Cancer	No c.
Cancer pred.	3	4998
No cancer pred.	2	4997

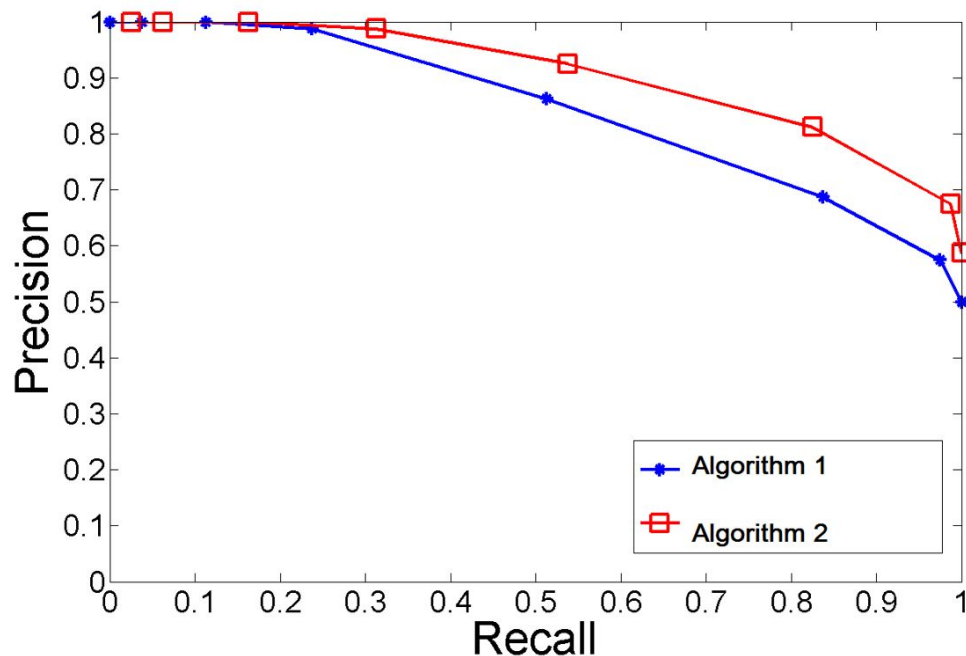
.5 accuracy

Never cancer predictor

	Cancer	No c.
Cancer pred.	0	0
No cancer pred.	5	9995

.999 accuracy

Area Under the Curve



Turning numeric prediction into classification with threshold ("operating point").

More Accuracy Measures for Classification Problems

- Lift
- Break even point
- F1 measure, etc
- Log loss (for class probabilities)
- Cohen's kappa, Gini coefficient (improvement over random)

Many Measures Beyond Classification

Regression:

- Mean Squared Error (MSE)
- Mean Absolute Percentage Error (MAPE)
- R^2 = percentage of variance explained by model

Rankings:

- Mean Average Precision in first K results (MAP@K)
- Mean Reciprocal Rank (MRR) (average rank for first correct prediction)
- Coverage (percentage of items ever recommended)
- Personalization (how similar predictions are for different users/queries)

Natural language processing:

- Translation and summarization -> comparing sequences (e.g ngrams) to human results with specialized metrics, e.g. [BLEU](#) and [ROUGE](#)
- Modeling text -> how well its probabilities match actual text, e.g., likelihood or [perplexity](#)

Always Compare Against Baselines!

Short Detour: Measurement

What is Measurement?

Measurement is the empirical, objective assignment of numbers, according to a rule derived from a model or theory, to attributes of objects or events with the intent of describing them. – Craner, Bond, “Software Engineering Metrics: What Do They Measure and How Do We Know?”

A quantitatively expressed reduction of uncertainty based on one or more observations. – Hubbard, “How to Measure Anything ...”

Examples: What/how do we measure?

Measuring model quality (cancer progn. model)

Measuring system quality (cancer progn. system)

Measuring user satisfaction (radiologists)

Measuring organizational goals (profits)

Measuring data scientist's competency (hiring)

Measuring software engineer's productivity (bonus)

Everything is Measurable

1. If X is something we care about, then X, by definition, must be detectable.
 - a. How could we care about things like “quality,” “risk,” “security,” or “public image” if these things were totally undetectable, directly or indirectly?
 - b. If we have reason to care about some unknown quantity, it is because we think it corresponds to desirable or undesirable results in some way.
2. If X is detectable, then it must be detectable in some amount.
 1. If you can observe a thing at all, you can observe more of it or less of it
 2. If we can observe it in some amount, then it must be measurable.
 3. *But: Not every measure is precise, not every measure is cost effective*

On Terminology

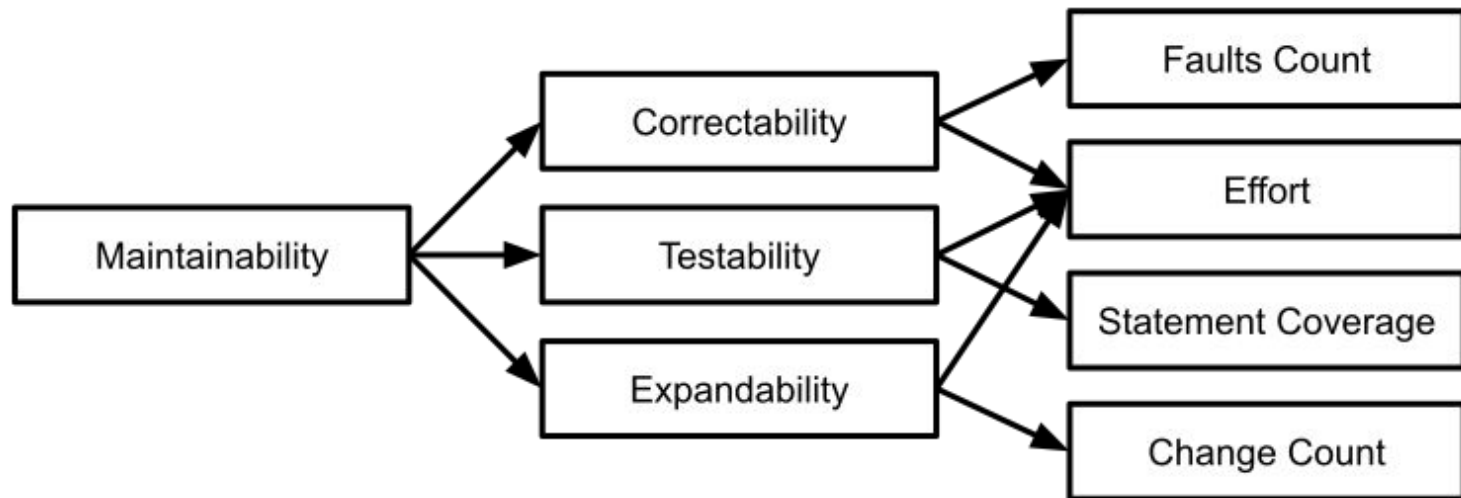


- **Quantification** is turning observations into numbers
- **Metric** and **measure** refer a method or standard format for measuring something (e.g., number of mistakes per hour)
 - Metric and measure synonymous for our purposes (some distinguish metrics as derived from multiple measures, or metrics to be standardizes measures)
- **Operationalization** is identifying and implementing a method to measure some factor (e.g., identifying mistakes from telemetry logs)

Composing/Decomposing Measures

Often higher-level measures are composed from lower level measures

Clear trace from specific low-level measurements to high-level metric



Stating Measures Precisely

- Always be precise about what we want to capture!
 - "measure accuracy" -> "evaluate accuracy with MAPE"
 - "evaluate test quality" -> "measure branch coverage with Jacoco"
 - "measure execution time" -> "average and 90%-quantile response time for REST-API x under normal load"
 - "assess developer skills" -> "measure average lines of code produced per day and number of bugs reported on code produced by that developer"
 - "measure customer happiness" -> "report response rate and average customer rating on survey shown to 5% of all customers (randomly selected)"
- Ideally: An independent party should be able to independently set up infrastructure to measure outcomes

Measuring

Make measurement clear and unambiguous. Ideally, third party can measure independently based on description.

Three steps:

1. **Measure:** What do we try to capture?
2. **Data collection:** What data is collected and how?
3. **Operationalization:** How is the measure computed from the data?

(Possible to repeat recursively when composing measures)

How to measure user satisfaction?



Three-Step Example

How to measure user satisfaction?

1. **Measure:** User satisfaction (scale 1-5)
2. **Data collection:** Show popup "How would you rate the diagnostic support today?" with 1-5 stars to 2% of randomly selected radiologists per day.
3. **Operationalization:** Report average star rating of all responses received in last 7 days.

Use common sense: No need to write formula for "average". Star ratings are implicitly understood. Some metrics are very common and operationalization is implied (e.g., F1-score).

Breakout: Measurement

Describe measurement in three-step structure (measure, data collection, operationalization) for:

1. Model quality of the cancer prognosis model
2. A leading indicator for the organizational goal of the company building the cancer prognosis system

As a group, post your answer to #1lecture tagging all group members.

Measurement Validity

Construct validity: Are we measuring what we intended to measure?

- Does the abstract concept match the specific scale/measurement used?
- e.g., IQ: What is it actually measuring?
- Other examples: Pain, language proficiency, personality...

Predictive validity: The extent to which the measurement can be used to explain some other characteristic of the entity being measured

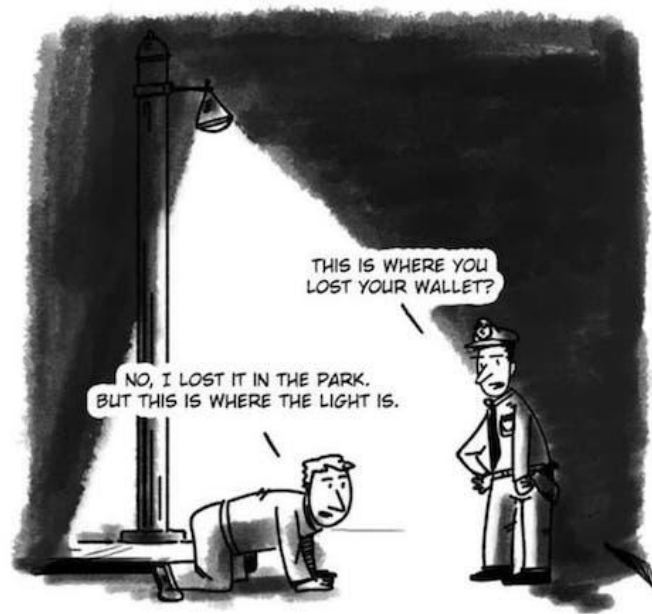
- e.g., Higher SAT scores => higher academic excellence?

External validity: Concerns the generalization of the findings to contexts and environments, other than the one studied

- e.g., Drug effectiveness on a test group: Does it hold over the general public?

Streetlight effect

- A type of *observational bias*
- People tend to look for something where it's easiest to do so
 - Use cheap proxy metrics that only poorly correlate with goal
 - e.g., number of daily active users as a measure of projected revenue



Risks with Measurements

Bad statistics: A basic misunderstanding of measurement theory and what is being measured.

Bad decisions: The incorrect use of measurement data, leading to unintended side effects.

Bad incentives: Disregard for the human factors, or how the cultural change of taking measurements will affect people.

Measuring Generalization

The Legend of the Failed Tank Detector



Overfitting in Cancer Prognosis?



Concept check: Overfit, Underfit, Train/Validation/Test Split, parameter, hyperparameter?

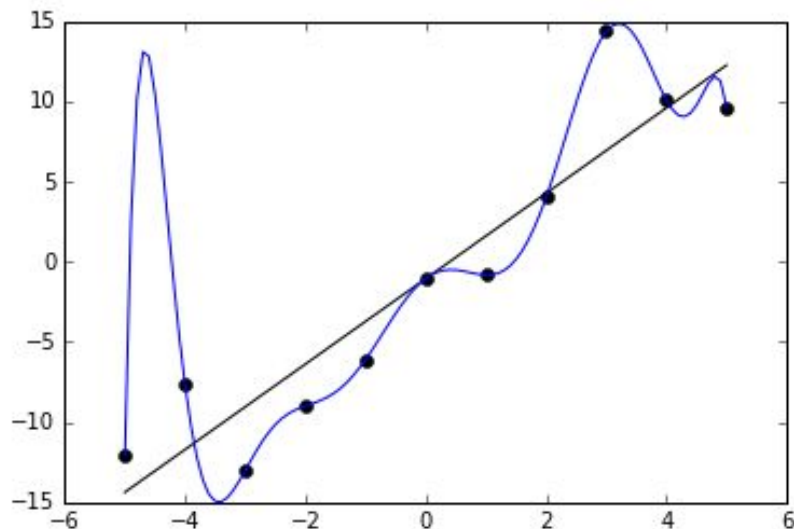


Overfitting/Underfitting

Overfitting: Model learned exactly for the input data, but does not generalize to unseen data (e.g., exact memorization)

Underfitting: Model makes very general observations but poorly fits to data (e.g., brightness in picture)

Typically adjust degrees of freedom during model learning to balance between overfitting and underfitting: can better learn the training data with more freedom (more complex models); but with too much freedom, will memorize details of the training data rather than generalizing



(CC SA 4.0 by [Ghiles](#))

Separate Training and Validation Data

Always test for generalization on *unseen* validation data
(independently sampled, typically from the same distribution)

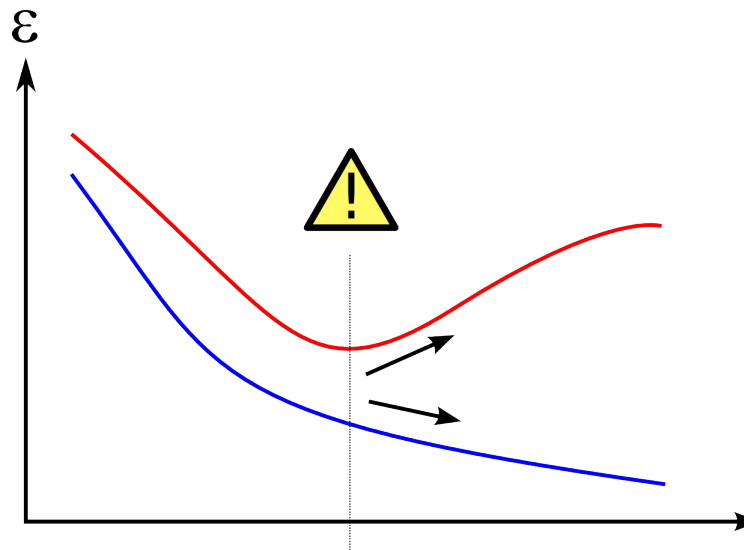
Accuracy on training data (or similar measure) used during learning
to find model parameters

```
train_xs, train_ys, valid_xs, valid_ys = split(all_xs, all_ys)
model = learn(train_xs, train_ys)

accuracy_train = accuracy(model, train_xs, train_ys)
accuracy_valid = accuracy(model, valid_xs, valid_ys)
```

accuracy_train >> *accuracy_valid* = sign of overfitting

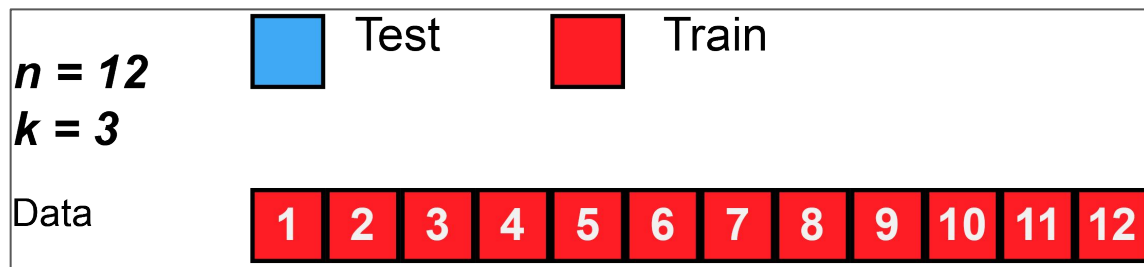
Detecting Overfitting



Change hyperparameter to
detect training accuracy
(blue)/validation accuracy (red) at
different degrees of freedom

Crossvalidation

- Motivation
 - Evaluate accuracy on different training and validation splits
 - Evaluate with small amounts of validation data
- Method: Repeated partitioning of data into train and validation data, train and evaluate model on each partition, average results
- Many split strategies, including
 - leave-one-out: evaluate on each datapoint using all other data for training
 - k-fold: k equal-sized partitions, evaluate on each training on others
 - repeated random sub-sampling (Monte Carlo)



Production Data -- The Ultimate Unseen Validation Data



more in a later lecture

Separate Training, Validation and Test Data

```
train_xs, train_ys, valid_xs, valid_ys, test_xs, test_ys =  
    split(all_xs, all_ys)  
  
best_model = null  
best_model_accuracy = 0  
for (hyperparameters in candidate_hyperparameters)  
    candidate_model = learn(train_xs, train_ys, hyperparameter)  
    model_accuracy = accuracy(model, valid_xs, valid_ys)  
    if (model_accuracy > best_model_accuracy)  
        best_model = candidate_model  
        best_model_accuracy = model_accuracy  
  
accuracy_test = accuracy(model, test_xs, test_ys)
```

Often a model is "tuned" manually or automatically on a validation set (hyperparameter optimization)

In this case, we can overfit on the validation set, separate test set is needed for final evaluation

On Terminology



```
// max_depth and min_support are hyperparameters
def learn_decision_tree(data, max_depth, min_support): Model =
    ...
// A, B, C are model parameters of model f
def f(outlook, temperature, humidity, windy) =
    if A==outlook
        return B*temperature + C*windy > 10
```

- The decisions in a model (weights, coefficients) are called **model parameter** of the model, their values are usually learned from the data
 - To a software engineer, these are **constants** in the learned function
- The inputs to the learning algorithm that are not the data are called **model hyperparameters**
 - To a software engineer, these are **parameters** to the learning algorithm, similar to compiler options

Common Pitfalls of Evaluating Model Quality

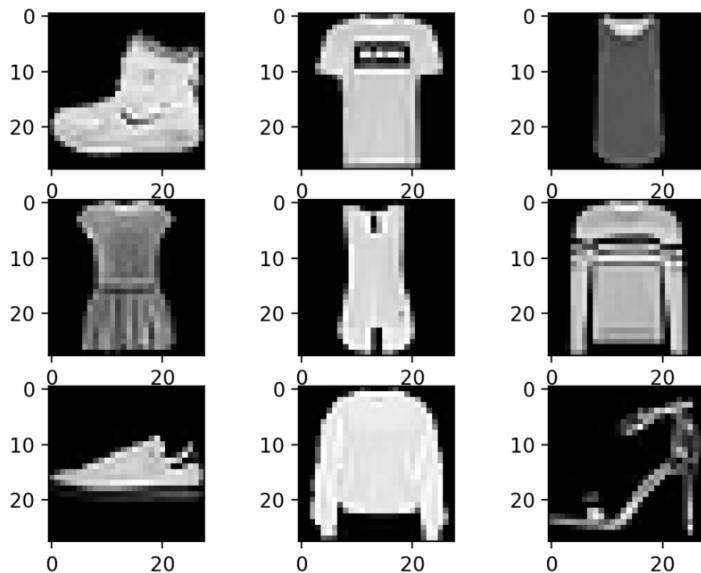
Common Pitfalls of Evaluating Model Quality?



0. Using Misleading Quality Measures

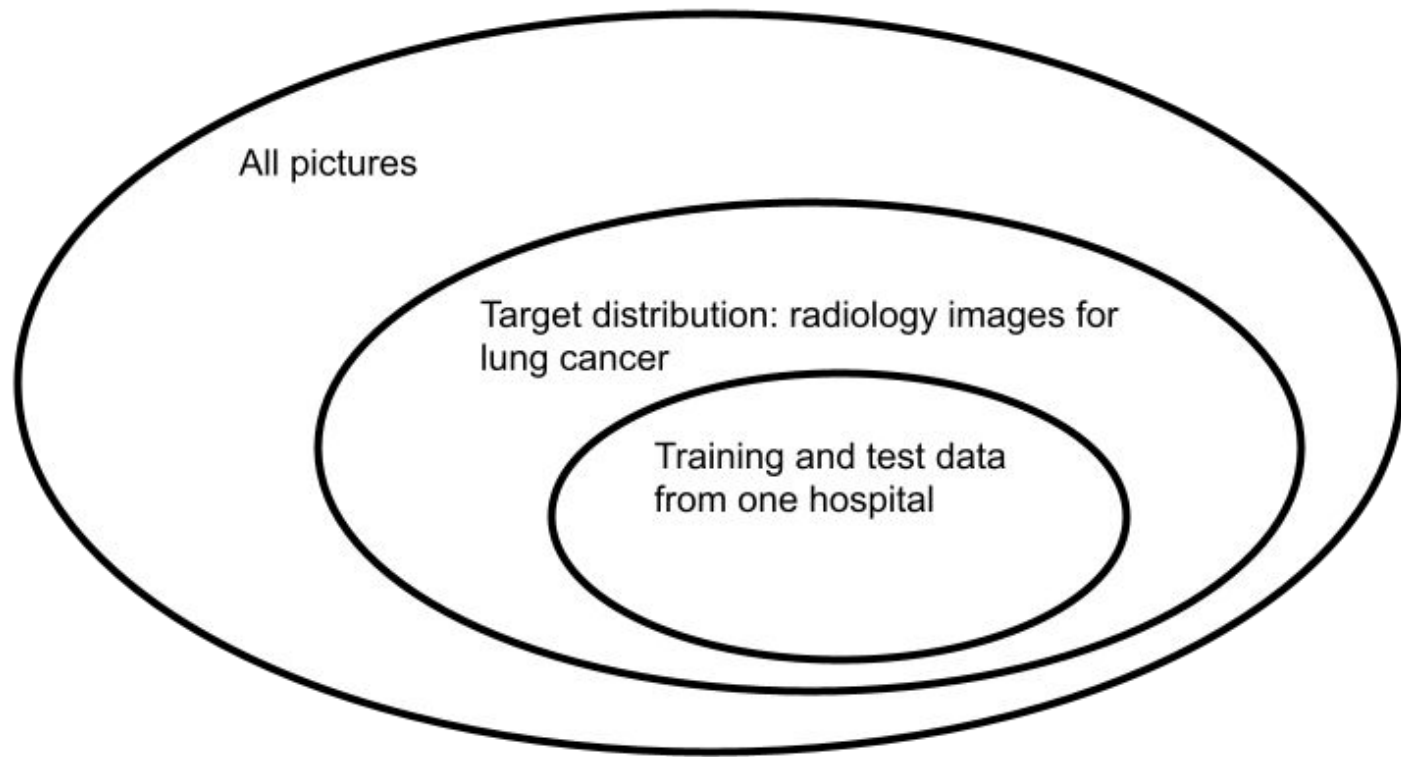
- using accuracy, when false positives are more harmful than false negatives
- comparing area under the curve, rather than relevant thresholds
- averaging over all populations, ignoring different results for subpopulations or different risks for certain predictions
- accuracy results on old static test data, when production data has shifted
- results on tiny validation sets
- reporting results without baseline
- ...

1. Data not Representative

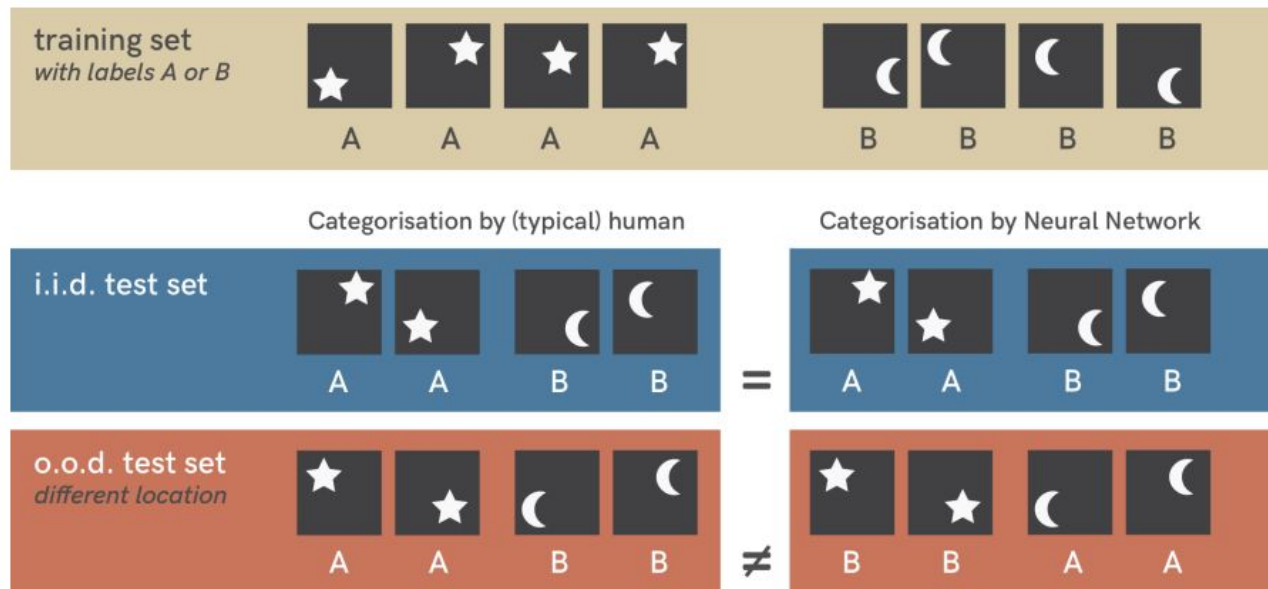


Often neither training nor test data representative of production data

1. Data not Representative



2. Shortcut Learning



"Spurious Correlations": Variables are associated but not causally related, due to coincidence or the presence of a certain third, unseen factor.

Figure from: Geirhos, Robert, et al. "[Shortcut learning in deep neural networks](#)." Nature Machine Intelligence 2, no. 11 (2020): 665-673.

Shortcut Learning



Caused by non-representative data, to some extent.

NeuralTalk2: A flock of birds flying in the air
Microsoft Azure: A group of giraffe standing next to a tree
Image: Fred Dunn, <https://www.flickr.com/photos/gratapictures> - CC-BY-NC

Generalization Beyond Training Data: Is this even fair to ask?



Generalization Beyond Training Data: What is "Distribution" for Foundation Models?



Representative Test Data in Practice?

Target distribution may **not be known** in early stages of the project

Target distribution may **shift over time** (“distribution drift”)

Evaluate the quality and representativeness of your test data (How?)

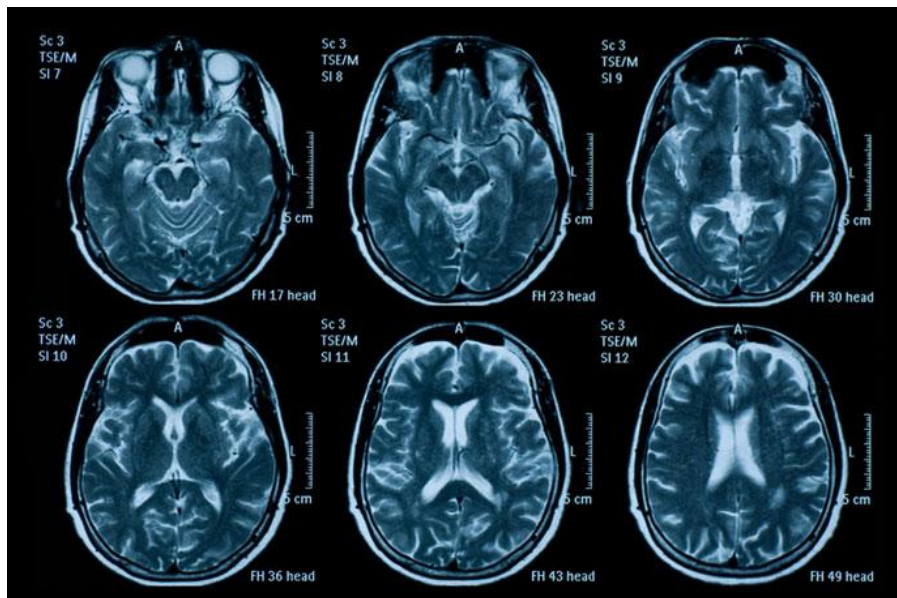
Production data is good test data, **but unlabeled...**

Monitoring and continuous data collection important! More later

3. Label Leakage



3. Label Leakage



Models have information they should not have access to!!

3. Label Leakage

Label or close correlates included in inputs

Examples:

- Input "interview conducted" in turnover prediction encodes human judgement
- Input "has bank account" associates with predicting whether somebody will open one

Is this a problem or a good thing?

Be cautious of "too good to be true" results

```
wordsVectorizer = CountVectorizer().fit(text)
wordsVector = wordsVectorizer.transform(text)
invTransformer = TfidfTransformer().fit(wordsVector)
invFreqOfWords = invTransformer.transform(wordsVector)
X = pd.DataFrame(invFreqOfWords.toarray())

train, test, spamLabelTrain, spamLabelTest =
    train_test_split(X, y, test_size = 0.5)
predictAndReport(train = train, test = test)
```

4. Data Leakage: Learning from Test Data

Test data *leaks* into training data

- surprisingly common in practice
- by accident, incorrect split -- or intentional using all data for training
- overlap between multiple datasets used
- data preprocessing on entire dataset
- tuning on validation data (e.g., cross-validation) without separate testing data

Results in overfitting and misleading accuracy measures

5. Data Leakage -- Overfitting in Continuous Experimentation

- Test data should be used exactly once – danger of overfitting with reuse
- Use of test sets to compare (hyperparameter-tuned) models in dashboards ➤ danger of overfitting
- Need fresh test data regularly
- Statistical techniques to approximate the needed amount of test data and the needed rotation

mlflow

GitHub

Docs

Listing Price Prediction

Experiment ID: 0

Artifact Location: /Users/matei/mlflow/demo/mlruns/0

Search Runs:

metrics.R2 > 0.24

Search

Filter Params:

alpha, lr

Filter Metrics:

rmse, r2

Clear

4 matching runs

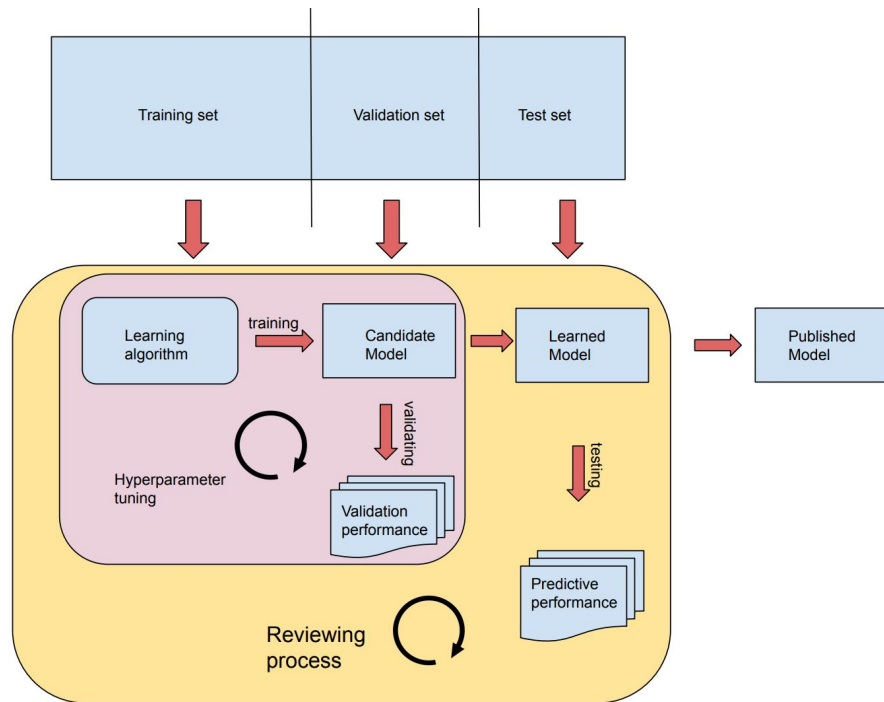
Compare Selected

Download CSV

	Time	User	Source	Version	Parameters		Metrics		
					alpha	l1_ratio	MAE	R2	RMSE
☐	17:37	matei	linear.py	3a1995	0.5	0.2	84.27	0.277	158.1
☐	17:37	matei	linear.py	3a1995	0.2	0.5	84.08	0.264	159.6
☐	17:37	matei	linear.py	3a1995	0.5	0.5	84.12	0.272	158.6
☐	17:37	matei	linear.py	3a1995	0	0	84.49	0.249	161.2

Recommended reading: Renggli, Cedric, Bojan Karlaš, Bolin Ding, Feng Liu, Kevin Schawinski, Wentao Wu, and Ce Zhang. "[Continuous integration of machine learning models with ease.ml/ci: Towards a rigorous yet practical treatment.](#)" arXiv preprint arXiv:1903.00278 (2019).

5. Data Leakage -- Overfit on Benchmarks



If many researchers publish best results on the same benchmark, collectively they perform "hyperparameter optimization" on the test set

(Figure by Andrea Passerini)

6. Independence of Data: e.g., Temporal

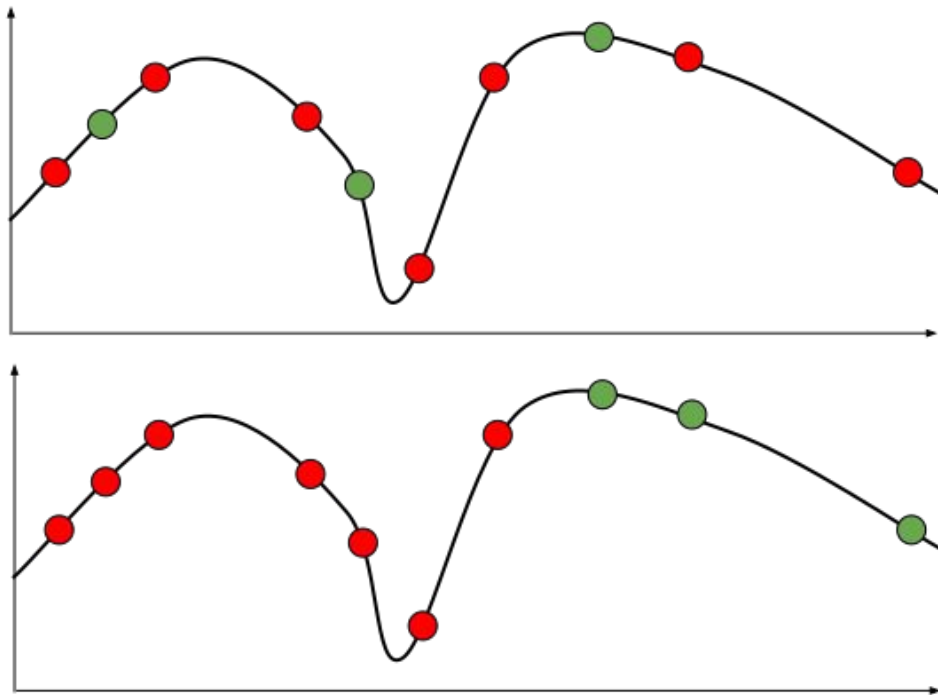
Attempt to predict the stock price development for different companies based on Twitter posts

Data: stock prices of 1000 companies over 4 years and Twitter mentions of those companies

Problems of random train--validation split?



6. Independence of Data: e.g., Temporal



Independence of Data: Related Datapoints

Example: Kaggle competition on detecting distracted drivers



Relation of datapoints may not be in the data (e.g., driver)

<https://www.fast.ai/2017/11/13/validation-sets/>

Data Dependence in Cancer Case Study?



Preliminary Summary: Common Pitfalls

- Always question the i.i.d. assumption
- Test data not representative
- Dependence between training and test data
- Misleading accuracy metrics
- Evaluating on training or validation data
- Label leakage
- Overfitting on test data through repeated evaluations

How to avoid? Ensure as part of process?

Note: i.i.d. = independent and identically distributed